

Name: Salman Khan

Sap Id: 73930

Subject: Programming
Assignment No-4

String:

Definition:-

A string is a data type in C++ that is used to store a sequence of characters.

C++ provides two types of string.

- 1) C style string (character array).
- 2) C++ string class (string) library.

Explanation:-

A string is simply a collection of character written inside double quotes.

Example of string.

"Hello"

"Computer"

The C++ string class is easier to use because it allows.

- Any length of text.
- Easy input output.

Paragraph:

The C++ string function are used to perform operation like copying, comparing, finding the length of character arrays when we work with C-style string (character arrays). manual handling is difficult because they do not grow or change.

Example:

1) strcpy (function)

```
#include <iostream>
```

```
#include <string>
```

```
using namespace std;
```

```
int main() {
```

```
    char source[] = "Programming";
```

```
    char destination[70];
```

```
    strcpy(destination, source);
```

```
    cout << "Copied String: " << destination << endl;
```

```
    return 0;
```

```
}
```

OUT PUT: Copied String: Programming

2) strcat () Function

```
#include <iostream>
#include <string>
using namespace std;
int main () {
    char str1[20] = "Hello";
    char str2[] = "World";
    strcat (str1, str2);
    cout << "After strcat: " << str1 << endl;
    return 0; }
```

OUT PUT: After 'strcat: Hello World

3) strcmp () Function

```
#include <string>
using namespace std;
int main () {
    char str1[] = "CPONS PONS";
    char str2[] = "CPONS PONS";
    if (strcmp (str1, str2) == 0)
        cout << "String are equal" << endl;
    else
        cout << "String are not equal" << endl;
    return 0; }
```

OUT PUT :-

String are equal

4) strlen() Function

```
int main() {
    char str[] = "Algorithm";
    cout << "Length of String: " <<
    strlen(str) << endl;
    return 0; }
```

Out Put: Length of String: 9

5) strncat() Function

```
int main() {
    char str1[20] = "Data";
    char str2[] = "Structure";
    strncat(str1, str2, 5);
    cout << "After strncat: " << str1 << endl;
    return 0; }
```

Out put: After strncat: Data struc

6) substr() Function

```
#include <iostream>
#include <string>
using namespace std;
int main() {
    string name = "Computer Science";
    string sub = name.substr(0, 8);
    cout << "Subtracting " << sub << endl;
    return 0; }
```

Out. put: Subtracting Computer